



Barcode : 770450637 Exam Series : XII Examination - Summer-2026
Total Marks : 33 Subject Code / Name : - PHYSICS

Question-wise Marks Summary

Q.No.	Marks	Q.No.	Marks	Q.No.	Marks	Q.No.	Marks
01	0	20b_i	1	29_iii	0	32b_ii_III	0*
02	0	20b_ii	1	29_iv_aOR	0	33a_i_IOR	0.5
03	1	21a	0	29_iv_b	0*	33a_i_IIOR	0
04	1	21b	1	30_i	0	33a_ii_IOR	0.5
05	0	22a	1	30_ii	1	33a_ii_IIOR	0.5
06	0	22b	0	30_iii	0	33a_ii_IIIOR	0
07	0	23a	1.5	30_iv_aOR	NA	33b_i	NA*
08	0	23b	1.5	30_iv_b	0*	33b_ii	NA*
09	1	24	3	31a_iOR	0.5		
10	1	25a	1.5	31a_iiOR	2		
11	0	25b	0	31b_i	NA*		
12	1	26aOR	NA	31b_ii_I	NA*		
13	1	26b	0*	31b_ii_II	NA*		
14	1	26b_i	0*	31b_ii_III	NA*		
15	0	26b_ii	0*	32a_i_IOR	NA		
16	0	27	3	32a_i_IIOR	NA		
17	2	28a	1	32a_iiOR	NA		
18	1.5	28b	0	32b_i	0*		
19	0.5	29_i	1	32b_ii_I	0*		
20aOR	NA*	29_ii	0	32b_ii_II	0*		

Note : ** marked marks are omitted for totaling.

केन्द्रीय माध्यमिक शिक्षा बोर्ड, दिल्ली
सीनियर स्कूल सर्टिफिकेट परीक्षा (कक्षा बारहवीं)
परीक्षाधी प्रवेश-यत्र के अनुसार करें

Each letter be written in one box and one box be left blank between each part of the name. In case Candidate's Name exceeds 22 letters, write first 22 letters.

कार्यालय उपयोग के लिए
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CHK : 7

मध्यमिक शिक्षा बोर्ड, दिल्ली
RD OF SECONDARY EDUCATION, DELHI
(HIGHER CERTIFICATE EXAMINATION (CLASS XII))
यार स्कूल सर्टिफिकेट परीक्षा (कक्षा बारहवीं)

05	06	07	08	09	10	TOTAL
15	16	17	18	19	20	TOTAL
25	26	27	28	29	30	TOTAL
35	36	37	38	39	40	TOTAL
GRAND TOTAL						

1. What is the purpose of the study?
The purpose of the study is to investigate the effect of the use of a mobile learning application on the learning outcomes of students in a mathematics course.

☐ **प्रमाणित करने वाले का प्रमाण** मैं 100% तक यकीन करता हूँ कि मैंने इस प्रश्नोत्तर पुस्तिका को ध्यान से और सही ढंग से जाँच लिया है।

☐ Also studied the answer book awarded 90% to 100% marks of maximum theory marks and found correct.

Steps: 1. 4 correct Equations of Examiners 2. 100% mark 100%

1. The first part of the document is a list of names and their corresponding addresses. The names are listed in the first column, and the addresses are listed in the second column. The names are: John Doe, Jane Smith, and Bob Johnson. The addresses are: 123 Main St, 456 Elm St, and 789 Oak St.

I certify that the contact marks have been uploaded on the app marks in January 2021 and downloaded and been supplied with coordinates and found correct.

Signature of Candidate (2) _____

6. Explain your results in the following questions. (10 points) (1) The rate of reaction was highest at 30°C. (2) The rate of reaction was lowest at 10°C. (3) The rate of reaction was intermediate at 20°C. (4) The rate of reaction was highest at 40°C. (5) The rate of reaction was lowest at 50°C. (6) The rate of reaction was intermediate at 60°C. (7) The rate of reaction was highest at 70°C. (8) The rate of reaction was lowest at 80°C. (9) The rate of reaction was intermediate at 90°C. (10) The rate of reaction was highest at 100°C.

I realize that I have checked this message twice and found that it still has many errors (except for being easier to correct, pointing to errors in the cover page is correct, leaving off most punctuation and letters). The original text is correct and the original author has been updated to the 12/20/2010/2011.

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Signature of Chair _____

CBSE

Instructions to Candidates

1. On receiving the answerbook,
 - (i) ensure that answer book contains 32 pages,
 - (ii) check that all pages are serially numbered (including title page),
 - (iii) fill in and blacken all the required details/fields correctly
 - (iv) use only blue-black or royal blue ink/gelballpoint pen
2. Write on each ruled line on both sides of the answer book.
3. Number your answers according to their numbers in the question paper.
4. Draw a line when a question (or a part thereof) is finished.
5. Do the rough work in the margin on right hand side of the page, which should be crossed out afterwards.
6. Securely tag your answer book with supplementary answer book(s), graphs, maps etc.
7. **DO NOT**
 - (i) waste pages by leaving wide margin,
 - (ii) make any special sign or mark in or outside the answer book, supplementary answer book, graph, map etc.
 - (iii) write your roll number, name of your school or place of examination in any of your answers.
 - (iv) fold the pages of the answer book.
 - (v) ask for supplementary answer book unless this answer book/previous supplementary answer book is finished.
 - (vi) leave the examination hall without handing over the answer book to the Assistant Superintendent.
 - (vii) Candidates are directed to write the Question Numbers only and only in the margin and not in the centre of the answer book or any other space in the answer book. This will not be evaluated.
 - (viii) Candidates are directed not to write or mark any answers or do any other rough work or any other work on the question paper other than writing their Roll number at the designated space.
8. Including in any of the following activities shall be deemed as use of unfair means, practice, result shall not be declared but marked as **UNFAIR MEANS (U/M)**.
 - (a) Having in possession any item or article which has been prohibited in examination centre or may be used for unfair practices including any stationary item, communication device, accessories, wearable items, instruments or any other material or information relevant or not relevant to the examination in the paper concerned.
 - (b) Paying /Placing someone else to write examination (impersonation) or candidate's behalf or preparing material for copying.
 - (c) Breaching examination Rules or any direction issued by CBSE from time to time, in connection with the conduct of EXAMINATIONS.
 - (d) Assisting other candidate to engage in malpractices, giving or receiving assistance directly or indirectly of any kind or attempting to do so.
 - (e) Writing questions or answers on any material other than the answer book given by the Centre Superintendent for writing purposes.
 - (f) Tearing of any page of the answer book or supplementary answer book etc.
 - (g) Contacting or communicating or trying to do so with any person, other than the Examination Staff, during the examination time in the examination Centre.
 - (h) Communicating with another candidate or the Assistant Superintendent directly or indirectly.
 - (i) Taking away the answer book out of the examination hall/centre.
 - (j) Smuggling out Question Paper or its part or smuggling out answer booklet/supplementary answer sheet or part thereof.
 - (k) Transferring any of the officials connected with the conduct of the examinations or threatening any of the candidates.
 - (l) Using or attempting to use any other undesirable method or means in connection with the examination.
 - (m) Forceful entry/exit in room / Examination Centre/hall.
 - (n) Use or attempted use of any electronic device after entering the examination centre.
 - (o) Uploading/sharing any examination related material, correct or wrong, on social media.
 - (p) Affixing/placing of fabricated photograph on the admit card.
 - (q) Erasing or obliterating any information printed on the ANSWER BOOK(s).
 - (r) Providing wrong information on the answer sheets.
 - (s) Having in possession question papers of previous years.
 - (t) If a candidate approaches any Authority/ies (person/s) related to the conduct of exams soliciting unauthorized privilege/s in these examinations.
 - (u) Taking legal counsel or any other to influence CBSE for gaining advantage in the favour, by providing false information.



Q.No	01	02	03	04	05	06	07	08	09	10	TOTAL
Marks											
Q.No	11	12	13	14	15	16	17	18	19	20	TOTAL
Marks											
Q.No	21	22	23	24	25	26	27	28	29	30	TOTAL
Marks											
Q.No	31	32	33	34	35	36	37	38	39	40	TOTAL
Marks											
TOTAL MARKS IN WORDS									GRAND TOTAL		



CBSE

Il risultato è che, se si considera l'intera popolazione, la media dei punteggi è di 100. Se si considera invece solo la popolazione che ha risposto "sì", la media dei punteggi è di 120. Se si considera invece solo la popolazione che ha risposto "no", la media dei punteggi è di 80. Questo è un esempio di bias di selezione.

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Signature of Applicant _____

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1. *આવેશ* અને *અવેશ* એ બે જુદા જુદા અર્થમાં વાપરવામાં આવે છે. *આવેશ* એ કોઈ કોઈ વસ્તુ સાથે અત્યંત પ્રત્યક્ષ સંપર્કમાં આવવાથી થતો હોય છે, જ્યારે *અવેશ* એ કોઈ કોઈ વસ્તુ સાથે અત્યંત પ્રત્યક્ષ સંપર્કમાં આવવાથી થતો હોય છે.

[illegible]

memory & emotion: Role of Environment

[illegible]

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TABLE 1. Continued

॥ प्रथमः पाठः ॥ ॐ नमो भगवते वासुदेवाय ॥ अथ श्रीभक्तिसहितप्रज्ञापादः ॥

☐ अधिकांश बट्टी जमीन का प्रयोग 10000 अंक प्राप्त पुलिसवालों की भाग की और सभी जगह

I certify that I have checked this answer book thoroughly. If a question is not correct, there is no immediate penalty. In the answer book, I have been consistent with the correct set of questions, justice and already in the answer.

☐ Also checked the present book awarded 90% to 100% marks of maximum theory marks and found correct.

Figure 3. 3D surface plot of the response surface.

1. *What is the main purpose of the study?*

I verify that the names have been updated in the and marks on pages 1 and 2 are correct and have compared with information and found correct.

It is important to note that the above information is for informational purposes only and should not be used as a basis for investment decisions. The information is subject to change without notice and is not intended to constitute an offer or solicitation of securities.

Family first names identified this summer have been listed first. Names not listed are considered still unknown. A complete listing of names on the order page is available. Additional names have been added to the group's book list and the current cards have been updated by the Commission.

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under the word $\mathcal{G}$ in the definition of the $\mathcal{G}$ -family, given $\mathcal{G} \in \mathcal{G}$ and $\mathcal{G} \in \mathcal{G}$ the family $\mathcal{G}$ is a $\mathcal{G}$ -family.

☐ Yes ☐ No

Figure 1

www.elsevier.com/locate/jmb

□ □ □

and then return to the original position.

2

Space for writing
Question Number

Section A

1. $B = B_0 \sin(106 - 4\pi)T$

C) $3.5 \times 10^7 \text{ ms}^{-1}$



0 01S1

2. A) Same wave length



0 02S1

3. D) 125Ω C) 100Ω



1 03S1

4. $\Phi = B \cdot A$

B) 400 m^2



1 04S1

5. $\theta = PE \sin \theta$

$W = \theta \times ds$

$\theta = W$

D) 2.29×10^{-26}



0 05S1

6) D) $2R$

$u = 1$



0 06S1

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$u = d = w$

$2 = 3 = w$

$u = d = 0$

$20 = 2L + 3L$

0.2

$\frac{2}{R}$

$u = IR$

$R = \frac{u}{I} = \frac{0.2}{I \times 10}$

Space for writing
Question Number

7) A) curves 1 & 3 both correspond to forward biasing of P-N junction. ☒

0 07S1

8) A) 1 ☒

0 08S1

$$\theta_{r2} = n^2 \theta_1$$

$$\begin{array}{r} 1 \\ 64 \\ 64 \\ \hline 128 \\ 256 \\ \hline 4096 \\ 159 \\ \hline 95 \\ 429 \end{array}$$

9) c) total internal reflection occurs for both green & blue lights.

✓ 1 09S1

$$\frac{h}{2\pi} = m v_p$$

10) D) $\frac{1}{3}$

✓ 1 10S1

$$m v \alpha = \frac{n h}{2\pi}$$

$$\begin{array}{r} 200 \times 10^9 \\ 1.65 \\ \hline 126 \end{array}$$

$$11) h\nu = \phi_0 + \frac{1}{2} m v^2$$

$$h\nu - \phi_0 = KE$$

B) 3.0 eV ☒

0 11S1

12) A) 2 n s

✓ 1 12S1



13) D) Both Assertion (A) & Reason (R) are false.

✓ 1 13S1

14) B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ^{not} the correct explanation of the Assertion (A).

✓ 1 14S1

15) C) Assertion (A) is true, but Reason (R) is false.

✗ 0 15S1

$$\frac{1}{2} = \frac{4}{8}$$

$$\frac{2}{8} = \frac{1}{2}$$

16) A) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

✗ 0 16S1

Section B

17) intrinsic semiconductor

- It is the semiconductor before doping
- $n_h = n_e$
no. of holes is equal to no. of e^- 's

extrinsic semiconductor

- It is the semiconductor after doping
- $n_h \neq n_e$
no. of holes is not equal to no. of e^-

✓ 1 17S1

17 : 1 + 1 = 2

✓ 1 17S2

Space for writing
Question Number

- 18) mass defect: A nucleus contains protons and neutrons so its mass should be the sum of masses of neutrons and protons in the nucleus but the actual mass is always less than that

✓ 0.5 18S1

$$\Delta m = n_p m_p + n_n m_n - M$$

✓ 0.5 18S3

Δm = sum of proton & neutron in the nucleus - original mass
where Δm is the mass defect

0 18S4

binding energy: is the minimum amount of energy required to break the nucleus into its constituent parts (i.e. protons & neutrons)

✓ 0.5 18S2

$$18 : 0.5 + 0.5 + 0.5 + 0 = 1.5$$

- 19) $\lambda = 5.0 \times 10^{-4}$
 $\nu = 6.6 \times 10^5$
 $\nu_0 = ?$
 $\nu_0 = \frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

energy, $E = h\nu$
 $E = h\nu = \frac{hc}{\lambda}$

$$h\nu_0 = \phi + KE$$

$$h\nu_0 = h(5.0 \times 10^{14}) + 6.6 \times 10^{-5}$$

$$h(\nu_0 - 5.0 \times 10^{14}) = 6.6 \times 10^{-5}$$

✗

0 19S2

✗

0 19S3

Ex 2
 $E = h\nu = \frac{hc}{\lambda}$
 $\lambda = \frac{hc}{E}$
 $\frac{h\nu}{e} = \frac{hc}{\lambda}$
 $\lambda = \frac{hc}{h\nu}$
 $\lambda = \frac{c}{\nu}$
 $\lambda = \frac{3 \times 10^8}{6.6 \times 10^5}$



$$h\nu_0 = h\nu_0 + KE$$

$$h\nu = h\nu_0 + KE$$

$$\boxed{\nu_0 = \frac{\nu - KE}{h}}$$

✓ 0.5 19S1



0 19S4

19 : 0.5 + 0 + 0 + 0 = 0.5

20) $E = 21$

$$\theta_1 = 3$$

$$I = 3$$

i) $E = V + IR$

$$E = I(R + \theta)$$

$$21 = 3(R + 3)$$

$$7 = R + 3$$

$$R = 4 \Omega$$

✓ 0.5 20b_is1

Resistance of the resistor is 4Ω

20b_i : 0.5 + 0.5 = 1

ii) $E = V + IR$

$$V = E - IR$$

$$V = 24 - 9$$

$$= 12 \text{ volts}$$

✓ 0.5 20b_iiS1

20b_ii : 0.5 + 0.5 = 1

terminal potential, 12 volts

✓ 0.5 20b_iiS2

21) a) for the bright fringe, phase difference

$$\phi = \frac{2\pi \times \Delta x}{\lambda}$$

$$\phi = \frac{2\pi \times n\lambda}{\lambda}$$

$$\text{Further, } \phi = \pi$$

$$\pi = 2\pi \times n$$

$$n = \frac{1}{2}$$



0 21aS1



0 21aS2

21a : 0 + 0 = 0

b) a) If one of the slit is closed we will not observe interference pattern but diffraction will happen. Since for interference we need two coherent sources. while for diffraction we need one slit only.

0.5 21bS1

0.5 21bS2

21b : 0.5 + 0.5 = 1

22) a) $E = 40 \sin(\omega t + \pi x)$

$B = 30 \sin(\omega t + \pi x)$



0 22aS1

Electromagnetic wave characteristics:
they are transverse in nature.



0.5 22aS2



$$22a : 0 + 0.5 + 0.5 = 1$$

- can propagate through vacuum
- propagate with their axes perpendicular to each other

✓ 0.5 22a S3

b)

$$\frac{1}{\sqrt{\epsilon_0 \mu_0}}$$



0 22b S1

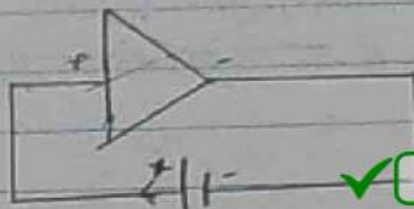
$$22b : 0 + 0 = 0$$

where ϵ_0 is the permittivity of electric field & μ_0 is the permeability of medium perpendicular of each other product gives velocity of EM wave in free space.

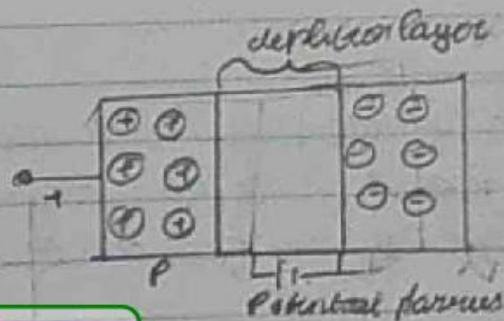
0 22b S2

24.

Forward Biasing



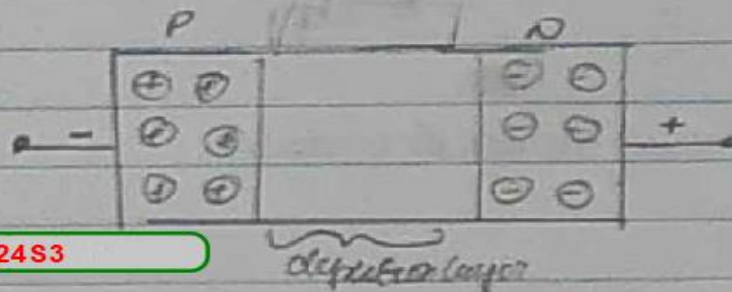
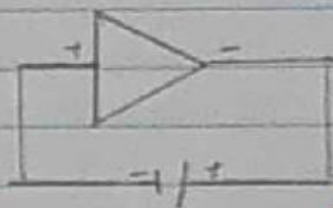
✓ 1 24 S1



when the positive side of the cell is connected to the positive side of the diode forward biasing takes place.
during forward biasing there is no attractive force to the depletion layer decreases & potential barrier decreases

✓ 0.5 24 S2

Reverse biasing



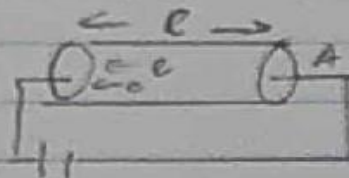
✓ 1 24S3

$$24 : 1 + 0.5 + 1 + 0.5 = 3$$

during reverse biasing the depletion layer increases here the positive side of diode is connected to the negative side of the P-N junction diode & vice versa here the potential barrier increases

✓ 0.5 24S4

- 25) a) consider a current carrying conductor having length l & area of cross section A if there are n no of electron then the no of electrons per unit volume is



volume of the conductor $V = A \times l$

total number of electron; $n = \frac{N}{V}$



$$N = nV$$

$$= nAl$$

Now the current flowing through the conductor

$$I = \frac{Nq}{t}$$

✓ 0.5 25a S1

$$= \frac{nAle}{t}$$

$$\text{where } \frac{l}{t} = vd$$

✓ 0.5 25a S2

$$25a : 0.5 + 0.5 + 0.5 = 1.5$$

$$I = v_d enA$$

✓ 0.5 25a S3

b) When length is doubled the drift velocity will also get doubled



0 25b S1

2b)

$$R = 30$$

$$\sigma = 8 \times 10^{-2}$$

$$I = 5A$$

$$B = 1T$$

$$\theta = 80^\circ$$

$$25b : 0 + 0 = 0$$

0 25b S2

Space for writing
Question Number

$$26) \theta = \frac{m v}{q B}$$

$$m = 6.4 \times 10^{-27}$$

$$q = 3.2 \times 10^{-19}$$

$$B = 0.5$$

$$\theta = \frac{6.4 \times 10^{-27} \times 5.0 \times 10^6}{0.5 \times 3.2 \times 10^{-19}}$$

$$= 0.4 \times 10^{-27+26+6} \times 8$$

$$= 3.2 \times 10^5 \text{ m}$$

0 26bS1

0 26bS2

0 26bS3

0 26bS4

$$26b : 0 + 0 + 0 + 0 = 0$$

ii) It will go undisturbed if the electric field & magnetic field is uniform

0 26b_iS1

i) If the electric field is $\perp$ to magnetic field

0 26b_iiS1

27) Faraday's law

1st law: A change in magnetic flux around a coil induces an emf in the coil

2nd law: The emf induced in a coil due to change is the change in magnetic flux per unit time

$$|e| = \frac{d\phi}{dt}$$

1 27S1



$$27 : 1 + 1 + 1 = 3$$

- emf can be produced by ~~induced~~ changing magnetic flux
- emf can be produced by changing current, which produces changing magnetic field lines further producing emf.

✓ 1 27S2

✓ 1 27S3

28) at distance of closest approach

$$KE = PE$$

$$\frac{1}{2}mv^2 = k \frac{q_1 q_2}{r_0^2} = k \frac{q_1 q_2}{r_0}$$

ii) for first particle

$$q_1 q_2 = \frac{k q_1 q_2}{\frac{1}{2} m_1 v_1^2}$$

for second particle

$$q_1 q_2 = \frac{k q_1 q_2}{\frac{1}{2} m_2 v_2^2}$$

iii) when they have same velocity

$$\frac{q_1 q_2}{q_1 q_2} = \frac{k q_1 q_2}{\frac{1}{2} m_1 v^2} \times \frac{\frac{1}{2} m_2 v^2}{k q_1 q_2}$$

✓ 0.5 28aS1

Space for writing
Question Number

$$\frac{q_1 q_1}{q_2} = \frac{m_2 q_1}{m_1 q_2}$$

✓ 0.5 28aS2

$$28a : 0.5 + 0.5 + 0 = 1$$

where q_1 & q_2 are charge of deuterium nucleus & α particle

0 28aS3

b) when the have same momentum

$$\frac{q_1 q_1}{q_2} = \frac{q_1 q_1}{q_2} \times \frac{1}{v_1} \times \frac{1}{v_2}$$

0 28bS1

$$\frac{q_1 q_1}{q_2} = \frac{q_1 v_2}{v_1 q_2} \times \frac{q_1 v_2}{v_1 q_2}$$

0 28bS3

0 28bS2

$$28b : 0 + 0 + 0 = 0$$

Section D

29) a) $\frac{1}{C_{net}} = \frac{1}{3} + \frac{1}{6}$

$$\frac{1}{C_{net}} = \frac{2+1}{6} = \frac{3}{6} \times \frac{1}{2}$$

$$C_{net} = 2 \mu F$$





i) D) 2 MF



1 29_iS1

ii) A) 0.8 A



0 29_iiS1

iii) D) 3V, 2V, 3V



0 29_iiiS1

iv) A) 6 MF



C) 12 MF

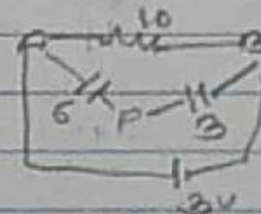
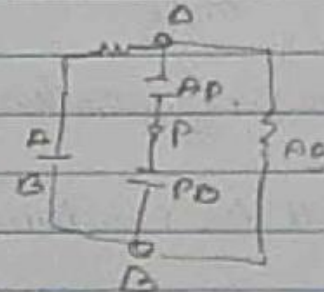
0 29_iv_aORS1

or

b) decreases

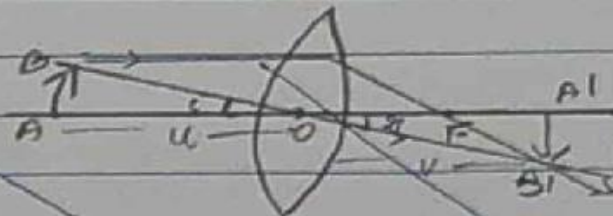


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$$\begin{aligned}
 q &= 6 \times 10^{-6} \text{ C} \\
 V &= 6 \text{ V} \\
 C &= \frac{q}{V} \\
 C &= \frac{6 \times 10^{-6}}{6} \\
 C &= 10^{-6} \text{ F} \\
 C &= 1 \text{ MF}
 \end{aligned}$$

31)



In $\triangle ABO$ & $\triangle A'B'O$
 $\angle AOB = \angle A'O B'$
 $\angle BAO = \angle B'A'O$
 $\therefore \triangle ABO \sim \triangle A'B'O$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$



Space for writing
Question Number

31) ii)

$$f = -20$$

$$u = -30$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

✓ 0.5 31a_iiORS1

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{30}$$

✓ 0.5 31a_iiORS2

$$\frac{1}{v} = \frac{30 - 20}{-600} = -\frac{10}{600}$$

$$v = -60 \text{ cm}$$

✓ 0.5 31a_iiORS3

The image is ~~not~~ Real, inverted

Position = 60 cm

✓ 0.5 31a_iiORS4

32) b)

Force exerted $F = \frac{40 \times 10^{-2} \times 2}{2 \times 10^{-2}}$
per unit lengthREPEAT
ANS+

$$\text{Given } \lambda = -2$$

$$-\lambda = \frac{q}{\epsilon} = -$$

$$\epsilon = \frac{q}{\lambda}$$

$$\frac{q}{\epsilon} = 3\lambda$$

$$\epsilon = \frac{q}{3\lambda}$$

0 32b_iS1

0 32b_iS2

31a_iiOR : 0.5 + 0.5 + 0.5 + 0.5 = 2



Force exerted on A due to P

$$= \frac{\mu_0 I_1 I_2 q}{2\pi a r} \quad \times$$

0 32b_iS3

Force exerted on B due to P

$$= \frac{\mu_0 I_1 I_2 q}{2\pi a r} - \frac{\mu_0 I_1 I_2}{2\pi a r} \quad \times$$

$$= \frac{\mu_0 I_1 I_2 q}{6\pi a r}$$

32b_i : 0 + 0 + 0 + 0 = 0

0 32b_iS4

The force applied are attractive same $\times$

33a_i_IOR : 0.5 + 0 = 0.5

33) I)

glow of light decreases, as some current is induced to the iron bar. $\times$

0.5 33a_i_IORS1

0 33a_i_IORS2

ii) The speed of the dark and bright glow increases decreases because frequency decrease means that alternating current change decreases. $\times$

0 33a_i_IORS1

33a_i_IOR : 0 + 0 = 0

ie. If the gas dark to glow change happen in 2 sec after decreasing frequency it will take more than 2 second. $\times$

0 33a_i_IORS2

Space for writing
Question Number

$$ii) V = V_0 \sin(\omega \cdot t)$$

$$V_0 = 280$$

$$\omega = 100\pi$$

$$R = 100 \Omega$$

$$C = 50 \mu F$$

here $(V_0)_0 > (V_0)_c$

$$i) Z = \sqrt{R^2 + (\omega C - \frac{1}{\omega C})^2}$$

✓ 0.5 33a_ii_IORS1

$$Z =$$

$$\omega C = \omega C$$

$$= 100\pi \times \frac{50}{1000} = 500 \Omega$$

$$33a_ii_IOR : 0.5 + 0 = 0.5$$

$$\omega C = \frac{1}{\omega C}$$

$$= \frac{1}{100\pi \times 50} = \frac{1}{50} = 0.02 \Omega$$

0 33a_ii_IORS2

$$Z = \sqrt{(160000) + (499.9)^2}$$



$$\frac{0.02}{100}$$

$$\frac{500.00}{0.02} = 499.99$$

$$Z = 100$$

$$ii) I_{rms} = \frac{V_0}{\sqrt{2}} = \frac{V_0}{R\sqrt{2}} = \frac{280}{100\sqrt{2}} = \frac{1}{2.0}$$

✓ 0.5 33a_ii_IORS1

$$33a_ii_IIOR : 0.5 + 0 = 0.5$$

0 33a_ii_IORS2



$$I_{\text{rms}} = 0.2$$

(u) $P = I_{\text{rms}} V_{\text{rms}} \cos \phi$
 $\cos \phi = \frac{I_{\text{rms}} V_{\text{rms}}}{P}$



0 33a_ii_IIIORS1

$$= \frac{0.2 \times 200}{P}$$

$$\cos \phi = \frac{40}{P}$$

$$\cos \phi = \frac{40 \times 200}{0.01 \times 0.02} = \frac{40}{0.0002} = \frac{4000}{14}$$

$$= \frac{200}{0.01} = \frac{20000}{7}$$



0 33a_ii_IIIORS2

$$\frac{200}{14}$$

$$\frac{147}{200} \cdot \frac{40}{20} = 10$$

33a_ii_IIIOR : 0 + 0 = 0

30) $m_1 = m$ $q_1 = -q$ K_1 for particle 1
 $m_2 = \frac{m}{2}$ $q_2 = +q$ K_2 for particle 2
 $K_1 = \frac{1}{2} m_1 v^2$



$$K_2 = \frac{1}{2} m_2 v^2$$



Space for writing
Question Number

$$\frac{K_1}{K_2} = \frac{\frac{1}{2} m_1 v^2}{\frac{1}{2} m_2 v^2}$$

$$= \frac{m_1}{m_2}$$

$$= \frac{1}{2} \times \frac{m_1 \times 2}{m_2}$$



$$P_1 = m_1 v$$

$$P_2 = m_2 v$$

$$P_1 = \frac{m_1 v}{2}$$

i) c) 4



0 30_iS1

ii) a) 2



1 30_iiS1

iii) c) both particles undergo uniform circular motion



0 30_iiiS1

iv) a) a) 2

or

v) b) $P_1 = 2P_2$



0 30_iv_bS1

$$P_1 = m_1 v$$

$$P_2 = \frac{m_2}{2} v$$

$$P_2 = \frac{P_1}{2}$$



23) a) $i = 45^\circ$

~~$D = 2i - A$~~ $A = 60^\circ$

$D = i + e - A$

$D = 90^\circ - 60^\circ$
 $= 30^\circ$

✓ 0.5 23aS1

23a : 0.5 + 0.5 + 0.5 = 1.5

Angle of minimum deviation

✓ 0.5 23aS2

$D = 30^\circ$

✓ 0.5 23aS3

b) $n_{21} = \frac{\sin(A + \frac{D}{2})}{\sin(\frac{A}{2})}$

✓ 0.5 23bS1

$= \frac{\sin \frac{90}{2}}{\sin \frac{60}{2}} = \frac{\sin 45}{\sin 30} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \frac{1}{\sqrt{2}} \times \frac{2}{1}$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2} \times \sqrt{2}}{1}$

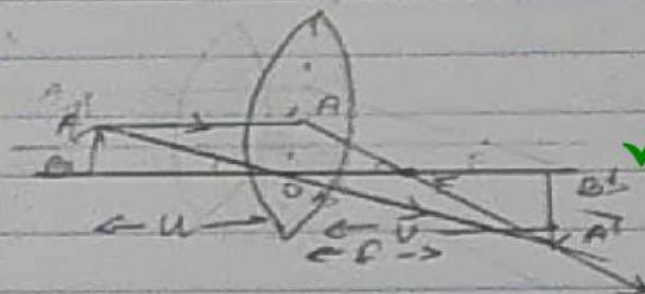
✓ 0.5 23bS2

$= \sqrt{2}$

23b : 0.5 + 0.5 + 0.5 = 1.5

Refractive index of the prism $n_{21} = \sqrt{2}$

✓ 0.5 23bS3

8) b
3) i)

✓ 0.5 31a_iORS1

In $\triangle AOB$ & $\triangle A'B'O$ $\angle O = \angle O$ (vertically opposite)

$$\therefore \frac{AB}{A'B'} = \frac{OB}{OB'} \quad (\triangle AOB \sim \triangle A'B'O)$$

✗

0 31a_iORS2

In $\triangle AOB$ & $\triangle A'B'O$

$$\frac{AO}{A'O} = \frac{OB}{OB'} \quad (\triangle AOB \sim \triangle A'B'O)$$

$$AO = A'O$$

✗

0 31a_iORS5

$$\frac{OB}{OB'} = \frac{OF}{OF'}$$

✗

0 31a_iORS3

$$-u = v - f$$

$$-u(v - f) = v(v - f)$$

dividing throughout

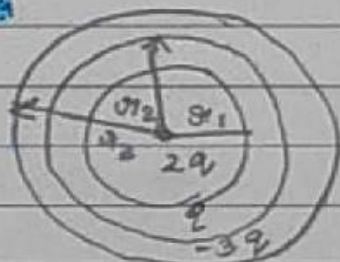
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

✗

0 31a_iORS4



32)b. (ii)


RPEAT
ANS+

according to Gauss's law
 $\oint \vec{E} \cdot d\vec{l} = \frac{1}{\epsilon_0} \times q_{\text{enc}}$

for a spherical shell : $E = \frac{q}{240}$

Electric flux, $\phi = E \cdot A$

$$= \frac{q}{240} \times 2\pi r^2 = \frac{q \pi r^2}{120}$$

for i) $r < r_1$
 $\frac{2q \pi r^2}{40}$

0 32b_ii_II S1

$$32b_ii_I : 0 + 0 = 0$$

ii) $\frac{q \pi r^2}{40}$

0 32b_ii_II S2

$$32b_ii_II : 0 + 0 = 0$$

III) i) $r > r_3$
 $\frac{-3q \pi r^2}{40}$

0 32b_ii_III S1

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Question Number

Surface charge density for a sphere

$$\sigma = \frac{Q}{A}$$

$$32b_ii_III : 0 + 0 = 0$$

$$\sigma = \frac{Q}{2\pi R^2}$$

$$0 \quad 32b_ii_IIIS2$$

i) for the sphere

$$\sigma = \frac{Q}{2\pi R^2}$$

for the shell (inner)

$$\sigma = \frac{-Q}{2\pi R_2^2}$$

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Question Number

25



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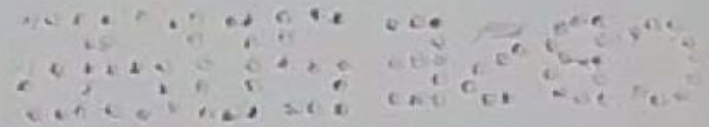
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